

Electrical Circuits

Lec 7: AC Circuits: Sinusoids and Phasors

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Book

- **Fundamentals of Electric Circuits**
 - Charles K. Alexander
 - Matthew N. O. Sadiku
 - 5th edition
- Chapter 9

Sinusoids

- A **sinusoid** is a signal that has the form of the sine or cosine function

Consider the sinusoidal voltage

$$v(t) = V_m \sin \omega t \quad (9.1)$$

where

V_m = the *amplitude* of the sinusoid

ω = the *angular frequency* in radians/s

ωt = the *argument* of the sinusoid

The sinusoid is shown in Fig. 9.1(a) as a function of its argument and in Fig. 9.1(b) as a function of time. It is evident that the sinusoid repeats itself every T seconds; thus, T is called the *period* of the sinusoid. From the two plots in Fig. 9.1, we observe that $\omega T = 2\pi$,

$$T = \frac{2\pi}{\omega}$$

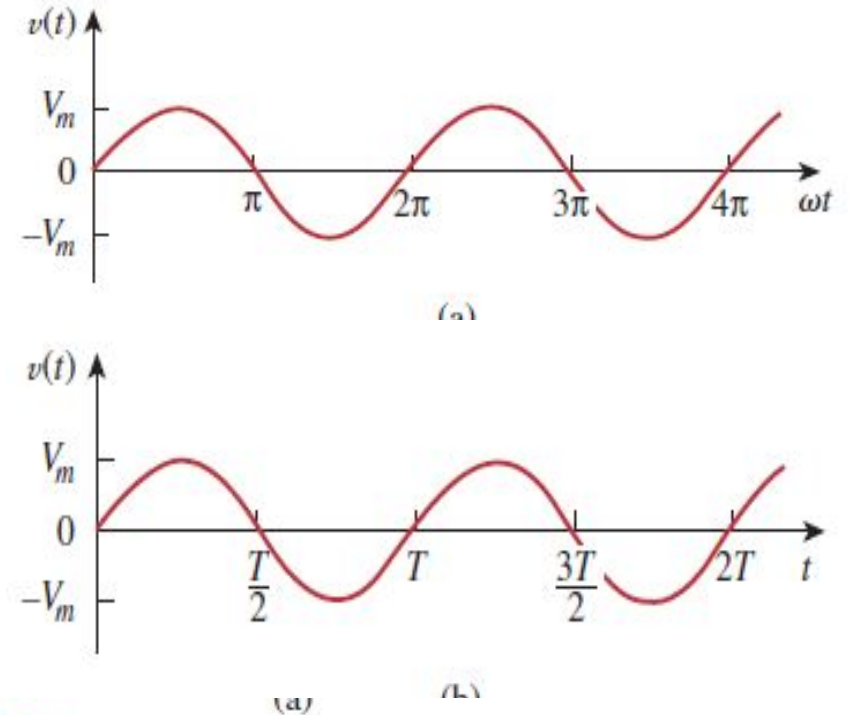


Figure 9.1

A sketch of $V_m \sin \omega t$: (a) as a function of ωt , (b) as a function of t .

The fact that $v(t)$ repeats itself every T seconds is shown by replacing t by $t + T$ in Eq. (9.1). We get

$$\begin{aligned}v(t + T) &= V_m \sin \omega(t + T) = V_m \sin \omega \left(t + \frac{2\pi}{\omega} \right) \\&= V_m \sin(\omega t + 2\pi) = V_m \sin \omega t = v(t)\end{aligned}\tag{9.3}$$

Hence,

$$v(t + T) = v(t)\tag{9.4}$$

that is, v has the same value at $t + T$ as it does at t and $v(t)$ is said to be *periodic*. In general,

A **periodic function** is one that satisfies $f(t) = f(t + nT)$, for all t and for all integers n .

Let us now consider a more general expression for the sinusoid,

$$v(t) = V_m \sin(\omega t + \phi) \quad (9.7)$$

where $(\omega t + \phi)$ is the argument and ϕ is the *phase*. Both argument and phase can be in radians or degrees.

Let us examine the two sinusoids

$$v_1(t) = V_m \sin \omega t \quad \text{and} \quad v_2(t) = V_m \sin(\omega t + \phi) \quad (9.8)$$

- The starting point of in Fig. 9.2 v_2 occurs first in time.
- Therefore, we say that v_2 *leads* v_1 by ϕ or that v_1 *lags* v_2 by ϕ .
- If $\phi \neq 0$, we also say that v_1 and v_2 are *out of phase*. If $\phi = 0$ then v_1 and v_2 are said to be *in phase*; they reach their minima and maxima at exactly the same time.

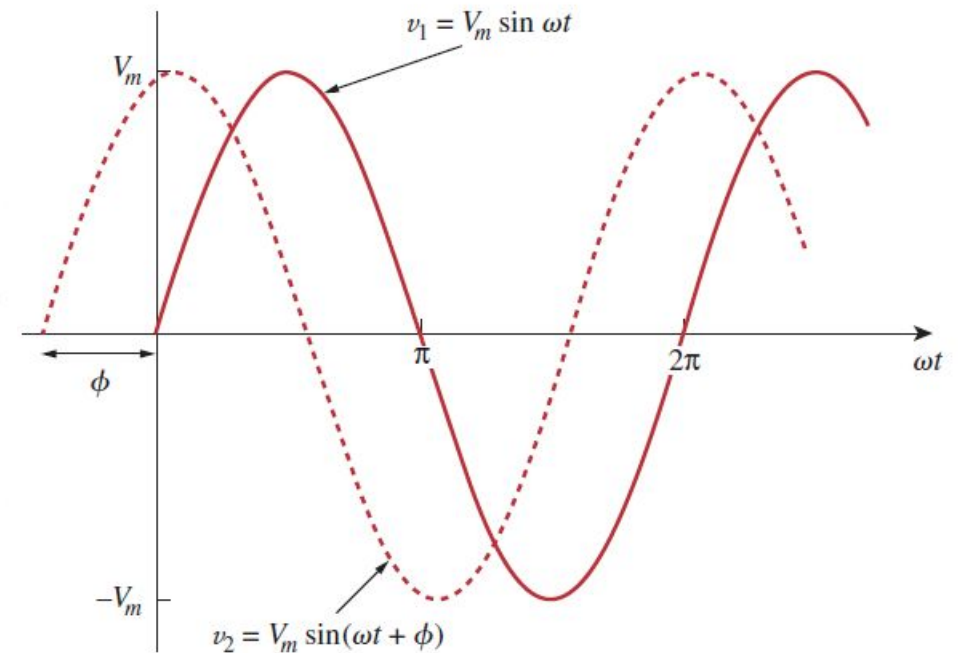


Figure 9.2

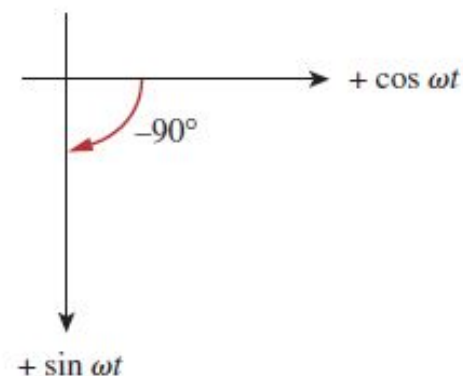
Two sinusoids with different phases.

$$\sin(\omega t \pm 180^\circ) = -\sin \omega t$$

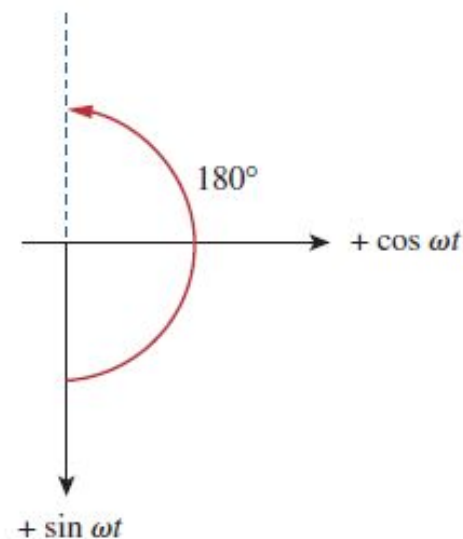
$$\cos(\omega t \pm 180^\circ) = -\cos \omega t$$

$$\sin(\omega t \pm 90^\circ) = \pm \cos \omega t$$

$$\cos(\omega t \pm 90^\circ) = \mp \sin \omega t$$



(a)



(b)

Figure 9.3

A graphical means of relating cosine and sine: (a) $\cos(\omega t - 90^\circ) = \sin \omega t$, (b) $\sin(\omega t + 180^\circ) = -\sin \omega t$.

$$A \cos \omega t + B \sin \omega t = C \cos(\omega t - \theta) \quad (9.11)$$

where

$$C = \sqrt{A^2 + B^2}, \quad \theta = \tan^{-1} \frac{B}{A} \quad (9.12)$$

$$3 \cos \omega t - 4 \sin \omega t = 5 \cos(\omega t + 53.1^\circ) \quad (9.13)$$

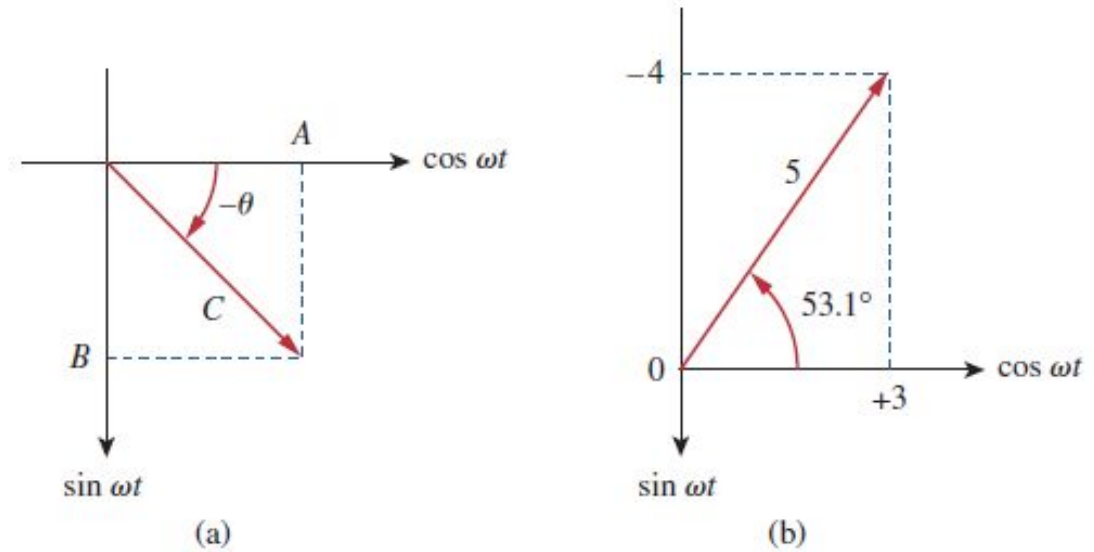


Figure 9.4

(a) Adding $A \cos \omega t$ and $B \sin \omega t$, (b) adding $3 \cos \omega t$ and $-4 \sin \omega t$.

Find the amplitude, phase, period, and frequency of the sinusoid

Example 9.1

$$v(t) = 12 \cos(50t + 10^\circ)$$

Solution:

The amplitude is $V_m = 12$ V.

The phase is $\phi = 10^\circ$.

The angular frequency is $\omega = 50$ rad/s.

The period $T = \frac{2\pi}{\omega} = \frac{2\pi}{50} = 0.1257$ s.

The frequency is $f = \frac{1}{T} = 7.958$ Hz.

Practice Problem 9.1

Given the sinusoid $30 \sin(4\pi t - 75^\circ)$, calculate its amplitude, phase, angular frequency, period, and frequency.

Answer: 30, -75° , 12.57 rad/s, 0.5 s, 2 Hz.

Calculate the phase angle between $v_1 = -10 \cos(\omega t + 50^\circ)$ and $v_2 = 12 \sin(\omega t - 10^\circ)$. State which sinusoid is leading.

Solution:

Let us calculate the phase in three ways. The first two methods use trigonometric identities, while the third method uses the graphical approach.

■ **METHOD 1** In order to compare v_1 and v_2 , we must express them in the same form. If we express them in cosine form with positive amplitudes,

$$\begin{aligned} v_1 &= -10 \cos(\omega t + 50^\circ) = 10 \cos(\omega t + 50^\circ - 180^\circ) \\ v_1 &= 10 \cos(\omega t - 130^\circ) \quad \text{or} \quad v_1 = 10 \cos(\omega t + 230^\circ) \end{aligned} \quad (9.2.1)$$

and

$$\begin{aligned} v_2 &= 12 \sin(\omega t - 10^\circ) = 12 \cos(\omega t - 10^\circ - 90^\circ) \\ v_2 &= 12 \cos(\omega t - 100^\circ) \end{aligned} \quad (9.2.2)$$

It can be deduced from Eqs. (9.2.1) and (9.2.2) that the phase difference between v_1 and v_2 is 30° . We can write v_2 as

$$v_2 = 12 \cos(\omega t - 130^\circ + 30^\circ) \quad \text{or} \quad v_2 = 12 \cos(\omega t + 260^\circ) \quad (9.2.3)$$

Comparing Eqs. (9.2.1) and (9.2.3) shows clearly that v_2 leads v_1 by 30° .

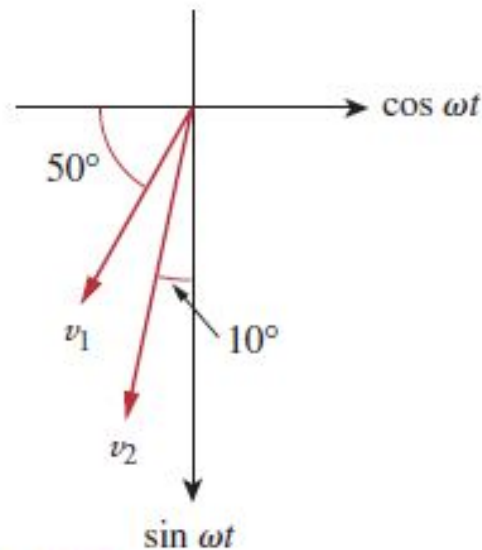


Figure 9.5

For Example 9.2.

■ **METHOD 2** Alternatively, we may express v_1 in sine form:

$$\begin{aligned} v_1 &= -10 \cos(\omega t + 50^\circ) = 10 \sin(\omega t + 50^\circ - 90^\circ) \\ &= 10 \sin(\omega t - 40^\circ) = 10 \sin(\omega t - 10^\circ - 30^\circ) \end{aligned}$$

But $v_2 = 12 \sin(\omega t - 10^\circ)$. Comparing the two shows that v_1 lags v_2 by 30° . This is the same as saying that v_2 leads v_1 by 30° .

■ **METHOD 3** We may regard v_1 as simply $-10 \cos \omega t$ with a phase shift of $+50^\circ$. Hence, v_1 is as shown in Fig. 9.5. Similarly, v_2 is $12 \sin \omega t$ with a phase shift of -10° , as shown in Fig. 9.5. It is easy to see from Fig. 9.5 that v_2 leads v_1 by 30° , that is, $90^\circ - 50^\circ - 10^\circ$.

Practice Problem 9.2

Find the phase angle between

$$i_1 = -4 \sin(377t + 55^\circ) \quad \text{and} \quad i_2 = 5 \cos(377t - 65^\circ)$$

Does i_1 lead or lag i_2 ?

Answer: 210° , i_1 leads i_2 .

Phasors

- Sinusoids are easily expressed in terms of phasors, which are more convenient to work with than sine and cosine functions.
- A phasor is a complex number that represents the amplitude and phase of a sinusoid.
- A complex number z can be written in rectangular form as where x is the real part of z ; y is the imaginary part of z .

$$z = x + jy \quad (9.14a)$$

- where x is the real part of z ; y is the imaginary part of z .

The complex number z can also be written in polar or exponential form as

$$z = r \angle \phi = re^{j\phi} \quad (9.14b)$$

where r is the magnitude of z , and ϕ is the phase of z . We notice that z can be represented in three ways:

$$\begin{array}{ll} z = x + jy & \text{Rectangular form} \\ z = r \angle \phi & \text{Polar form} \\ z = re^{j\phi} & \text{Exponential form} \end{array} \quad (9.15)$$

The relationship between the rectangular form and the polar form is shown in Fig. 9.6, where the x axis represents the real part and the y axis represents the imaginary part of a complex number. Given x and y , we can get r and ϕ as

$$r = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x} \quad (9.16a)$$

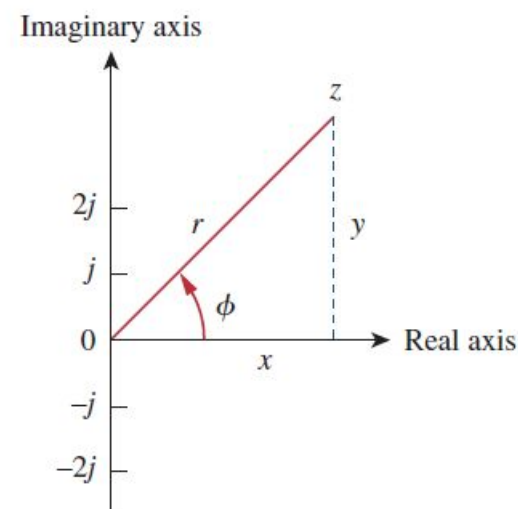


Figure 9.6
Representation of a complex number $z = x + jy = r \angle \phi$.

On the other hand, if we know r and ϕ , we can obtain x and y as

$$x = r \cos \phi, \quad y = r \sin \phi \quad (9.16b)$$

Thus, z may be written as

$$z = x + jy = r \angle \phi = r(\cos \phi + j \sin \phi) \quad (9.17)$$

Addition and subtraction of complex numbers are better performed in rectangular form; multiplication and division are better done in polar form. Given the complex numbers

$$z = x + jy = r \angle \phi, \quad z_1 = x_1 + jy_1 = r_1 \angle \phi_1$$
$$z_2 = x_2 + jy_2 = r_2 \angle \phi_2$$

Addition:

$$z_1 + z_2 = (x_1 + x_2) + j(y_1 + y_2) \quad (9.18a)$$

Subtraction:

$$z_1 - z_2 = (x_1 - x_2) + j(y_1 - y_2) \quad (9.18b)$$

Multiplication:

$$z_1 z_2 = r_1 r_2 \underline{\angle \phi_1 + \phi_2} \quad (9.18c)$$

Division:

$$\frac{z_1}{z_2} = \frac{r_1}{r_2} \underline{\angle \phi_1 - \phi_2} \quad (9.18d)$$

Reciprocal:

$$\frac{1}{z} = \frac{1}{r} \underline{\angle -\phi} \quad (9.18e)$$

Square Root:

$$\sqrt{z} = \sqrt{r} \angle \underline{\phi/2} \quad (9.18f)$$

Complex Conjugate:

$$z^* = x - jy = r \angle \underline{-\phi} = re^{-j\phi} \quad (9.18g)$$

Note that from Eq. (9.18e),

$$\frac{1}{j} = -j \quad (9.18h)$$

The idea of phasor representation is based on Euler's identity. In general,

$$e^{\pm j\phi} = \cos\phi \pm j \sin\phi \quad (9.19)$$

which shows that we may regard $\cos\phi$ and $\sin\phi$ as the real and imaginary parts of $e^{j\phi}$; we may write

$$\cos\phi = \operatorname{Re}(e^{j\phi}) \quad (9.20a)$$

$$\sin\phi = \operatorname{Im}(e^{j\phi}) \quad (9.20b)$$

where Re and Im stand for the *real part of* and the *imaginary part of*. Given a sinusoid $v(t) = V_m \cos(\omega t + \phi)$, we use Eq. (9.20a) to express $v(t)$ as

$$v(t) = V_m \cos(\omega t + \phi) = \operatorname{Re}(V_m e^{j(\omega t + \phi)}) \quad (9.21)$$

or

$$v(t) = \operatorname{Re}(V_m e^{j\phi} e^{j\omega t}) \quad (9.22)$$

Thus,

$$v(t) = \operatorname{Re}(\mathbf{V} e^{j\omega t}) \quad (9.23)$$

where

$$\mathbf{V} = V_m e^{j\phi} = V_m \angle \phi \quad (9.24)$$

Equations (9.21) through (9.23) reveal that to get the phasor corresponding to a sinusoid, we first express the sinusoid in the cosine form so that the sinusoid can be written as the real part of a complex number. Then we take out the time factor $e^{j\omega t}$, and whatever is left is the phasor corresponding to the sinusoid. By suppressing the time factor, we transform the sinusoid from the time domain to the phasor domain. This transformation is summarized as follows:

$$\begin{array}{ccc} v(t) = V_m \cos(\omega t + \phi) & \Leftrightarrow & \mathbf{V} = V_m \angle \phi \\ \text{(Time-domain representation)} & & \text{(Phasor-domain representation)} \end{array} \quad (9.25)$$

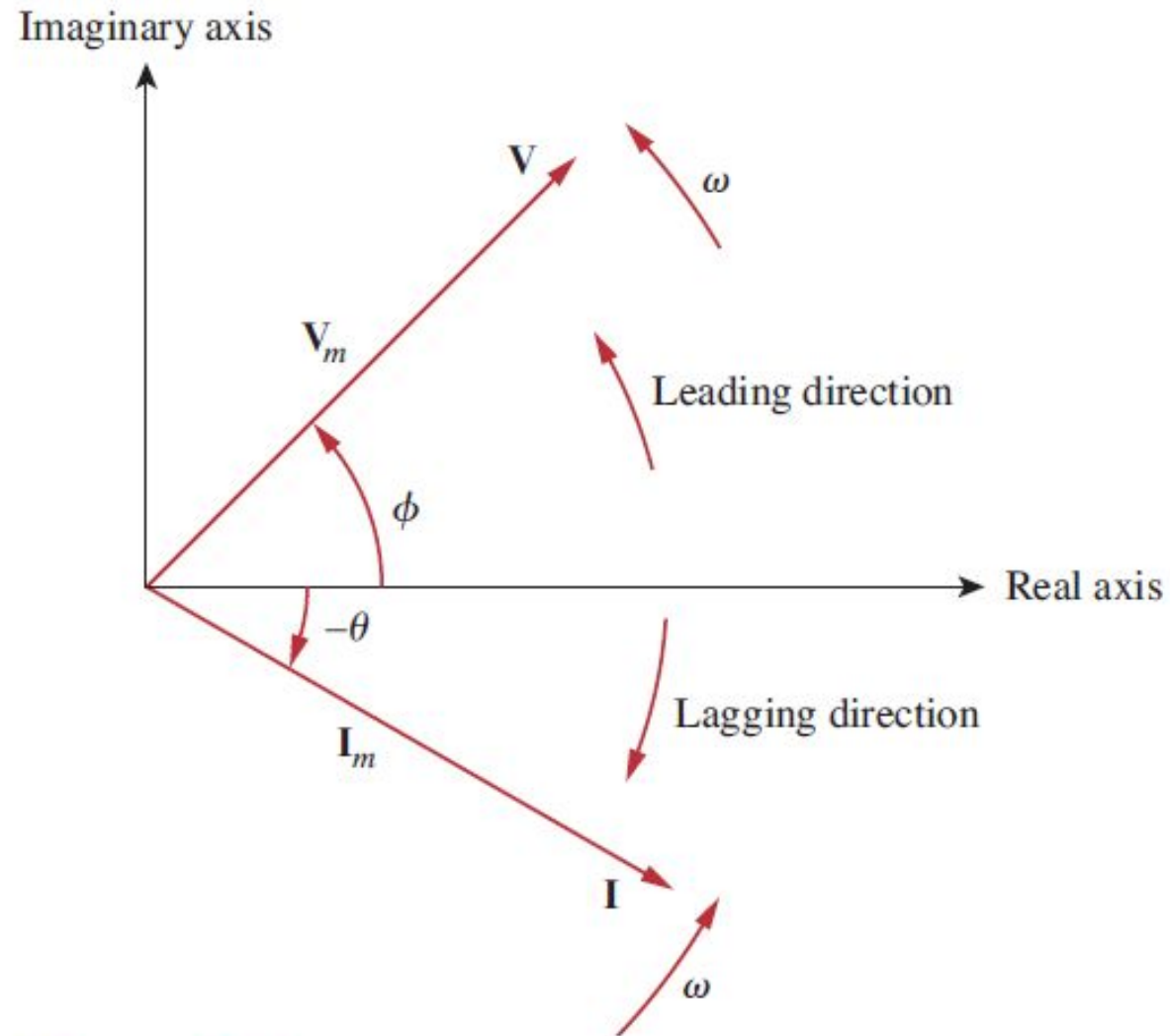


Figure 9.8

A phasor diagram showing $\mathbf{V} = V_m \angle \phi$ and $\mathbf{I} = I_m \angle -\theta$.

TABLE 9.1

Sinusoid-phasor transformation.

Time domain representation	Phasor domain representation
$V_m \cos(\omega t + \phi)$	$V_m \angle \phi$
$V_m \sin(\omega t + \phi)$	$V_m \angle \phi - 90^\circ$
$I_m \cos(\omega t + \theta)$	$I_m \angle \theta$
$I_m \sin(\omega t + \theta)$	$I_m \angle \theta - 90^\circ$

From Eqs. (9.23) and (9.24), $v(t) = \text{Re}(\mathbf{V}e^{j\omega t}) = V_m \cos(\omega t + \phi)$, so that

$$\begin{aligned}\frac{dv}{dt} &= -\omega V_m \sin(\omega t + \phi) = \omega V_m \cos(\omega t + \phi + 90^\circ) \\ &= \text{Re}(\omega V_m e^{j\omega t} e^{j\phi} e^{j90^\circ}) = \text{Re}(j\omega \mathbf{V} e^{j\omega t})\end{aligned}\tag{9.26}$$

This shows that the derivative $v(t)$ is transformed to the phasor domain as $j\omega\mathbf{V}$

$$\begin{array}{ccc} \frac{dv}{dt} & \Leftrightarrow & j\omega\mathbf{V} \\ \text{(Time domain)} & & \text{(Phasor domain)} \end{array} \quad (9.27)$$

Differentiating a sinusoid is equivalent to multiplying its corresponding phasor by $j\omega$.

Similarly, the integral of $v(t)$ is transformed to the phasor domain as $\mathbf{V}/j\omega$

$$\begin{array}{ccc} \int v \, dt & \Leftrightarrow & \frac{\mathbf{V}}{j\omega} \\ \text{(Time domain)} & & \text{(Phasor domain)} \end{array} \quad (9.28)$$

Integrating a sinusoid is equivalent to dividing its corresponding phasor by $j\omega$.

Example 9.3

Evaluate these complex numbers:

(a) $(40\angle 50^\circ + 20\angle -30^\circ)^{1/2}$

(b) $\frac{10\angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*}$

Solution:

(a) Using polar to rectangular transformation,

$$40\angle 50^\circ = 40(\cos 50^\circ + j \sin 50^\circ) = 25.71 + j30.64$$

$$20\angle -30^\circ = 20[\cos(-30^\circ) + j \sin(-30^\circ)] = 17.32 - j10$$

Adding them up gives

$$40\angle 50^\circ + 20\angle -30^\circ = 43.03 + j20.64 = 47.72\angle 25.63^\circ$$

Taking the square root of this,

$$(40\angle 50^\circ + 20\angle -30^\circ)^{1/2} = 6.91\angle 12.81^\circ$$

(b) Using polar-rectangular transformation, addition, multiplication, and division,

$$\begin{aligned}\frac{10\angle -30^\circ + (3 - j4)}{(2 + j4)(3 - j5)^*} &= \frac{8.66 - j5 + (3 - j4)}{(2 + j4)(3 + j5)} \\ &= \frac{11.66 - j9}{-14 + j22} = \frac{14.73\angle -37.66^\circ}{26.08\angle 122.47^\circ} \\ &= 0.565\angle -160.13^\circ\end{aligned}$$

Practice Problem 9.3

Evaluate the following complex numbers:

(a) $[(5 + j2)(-1 + j4) - 5\angle 60^\circ]^*$

(b) $\frac{10 + j5 + 3\angle 40^\circ}{-3 + j4} + 10\angle 30^\circ + j5$

Answer: (a) $-15.5 - j13.67$, (b) $8.293 + j7.2$.

Example 9.4

Transform these sinusoids to phasors:

(a) $i = 6 \cos(50t - 40^\circ) \text{ A}$

(b) $v = -4 \sin(30t + 50^\circ) \text{ V}$

Solution:

(a) $i = 6 \cos(50t - 40^\circ)$ has the phasor

$$\mathbf{I} = 6 \angle -40^\circ \text{ A}$$

(b) Since $-\sin A = \cos(A + 90^\circ)$,

$$\begin{aligned} v &= -4 \sin(30t + 50^\circ) = 4 \cos(30t + 50^\circ + 90^\circ) \\ &= 4 \cos(30t + 140^\circ) \text{ V} \end{aligned}$$

The phasor form of v is

$$\mathbf{V} = 4 \angle 140^\circ \text{ V}$$

Practice Problem 9.4

Express these sinusoids as phasors:

(a) $v = 7 \cos(2t + 40^\circ)$ V

(b) $i = -4 \sin(10t + 10^\circ)$ A

Answer: (a) $\mathbf{V} = 7 \angle 40^\circ$ V, (b) $\mathbf{I} = 4 \angle 100^\circ$ A.

Example 9.5

Find the sinusoids represented by these phasors:

(a) $\mathbf{I} = -3 + j4$ A

(b) $\mathbf{V} = j8e^{-j20^\circ}$ V

Solution:

(a) $\mathbf{I} = -3 + j4 = 5\angle 126.87^\circ$. Transforming this to the time domain gives

$$i(t) = 5 \cos(\omega t + 126.87^\circ) \text{ A}$$

(b) Since $j = 1\angle 90^\circ$,

$$\begin{aligned}\mathbf{V} &= j8\angle -20^\circ = (1\angle 90^\circ)(8\angle -20^\circ) \\ &= 8\angle 90^\circ - 20^\circ = 8\angle 70^\circ \text{ V}\end{aligned}$$

Converting this to the time domain gives

$$v(t) = 8 \cos(\omega t + 70^\circ) \text{ V}$$

Practice Problem 9.5

Find the sinusoids corresponding to these phasors:

(a) $\mathbf{V} = -25 \angle 40^\circ \text{ V}$

(b) $\mathbf{I} = j(12 - j5) \text{ A}$

Answer: (a) $v(t) = 25 \cos(\omega t - 140^\circ) \text{ V}$ or $25 \cos(\omega t + 220^\circ) \text{ V}$,
(b) $i(t) = 13 \cos(\omega t + 67.38^\circ) \text{ A}$.

Given $i_1(t) = 4 \cos(\omega t + 30^\circ)$ A and $i_2(t) = 5 \sin(\omega t - 20^\circ)$ A, find their sum.

Solution:

Here is an important use of phasors—for summing sinusoids of the same frequency. Current $i_1(t)$ is in the standard form. Its phasor is

$$\mathbf{I}_1 = 4 \angle 30^\circ$$

We need to express $i_2(t)$ in cosine form. The rule for converting sine to cosine is to subtract 90° . Hence,

$$i_2 = 5 \cos(\omega t - 20^\circ - 90^\circ) = 5 \cos(\omega t - 110^\circ)$$

and its phasor is

$$\mathbf{I}_2 = 5 \angle -110^\circ$$

If we let $i = i_1 + i_2$, then

$$\begin{aligned}\mathbf{I} &= \mathbf{I}_1 + \mathbf{I}_2 = 4 \angle 30^\circ + 5 \angle -110^\circ \\ &= 3.464 + j2 - 1.71 - j4.698 = 1.754 - j2.698 \\ &= 3.218 \angle -56.97^\circ \text{ A}\end{aligned}$$

Transforming this to the time domain, we get

$$i(t) = 3.218 \cos(\omega t - 56.97^\circ) \text{ A}$$

Of course, we can find $i_1 + i_2$ using Eq. (9.9), but that is the hard way.

Practice Problem 9.6

If $v_1 = -10 \sin(\omega t - 30^\circ)$ V and $v_2 = 20 \cos(\omega t + 45^\circ)$ V, find $v = v_1 + v_2$.

Answer: $v(t) = 29.77 \cos(\omega t + 49.98^\circ)$ V.

Example 9.7

Using the phasor approach, determine the current $i(t)$ in a circuit described by the integrodifferential equation

$$4i + 8 \int i dt - 3 \frac{di}{dt} = 50 \cos(2t + 75^\circ)$$

Solution:

We transform each term in the equation from time domain to phasor domain. Keeping Eqs. (9.27) and (9.28) in mind, we obtain the phasor form of the given equation as

$$4\mathbf{I} + \frac{8\mathbf{I}}{j\omega} - 3j\omega\mathbf{I} = 50\angle 75^\circ$$

But $\omega = 2$, so

$$\begin{aligned}\mathbf{I}(4 - j4 - j6) &= 50\angle 75^\circ \\ \mathbf{I} &= \frac{50\angle 75^\circ}{4 - j10} = \frac{50\angle 75^\circ}{10.77\angle -68.2^\circ} = 4.642\angle 143.2^\circ \text{ A}\end{aligned}$$

Converting this to the time domain,

$$i(t) = 4.642 \cos(2t + 143.2^\circ) \text{ A}$$

Keep in mind that this is only the steady-state solution, and it does not require knowing the initial values.

Practice Problem 9.7

Find the voltage $v(t)$ in a circuit described by the integrodifferential equation

$$2\frac{dv}{dt} + 5v + 10 \int v dt = 50 \cos(5t - 30^\circ)$$

using the phasor approach.

Answer: $v(t) = 5.3 \cos(5t - 88^\circ) \text{ V}$.

Phasor Relationships for Circuit Elements

- Now that we know how to represent a voltage or current in the phasor or frequency domain, one may legitimately ask how we apply this to circuits involving the passive elements R , L , and C .

We begin with the resistor. If the current through a resistor R is $i = I_m \cos(\omega t + \phi)$, the voltage across it is given by Ohm's law as

$$v = iR = RI_m \cos(\omega t + \phi) \quad (9.29)$$

The phasor form of this voltage is

$$\mathbf{V} = RI_m \angle \phi \quad (9.30)$$

But the phasor representation of the current is $\mathbf{I} = I_m \angle \phi$. Hence,

$$\mathbf{V} = R\mathbf{I} \quad (9.31)$$

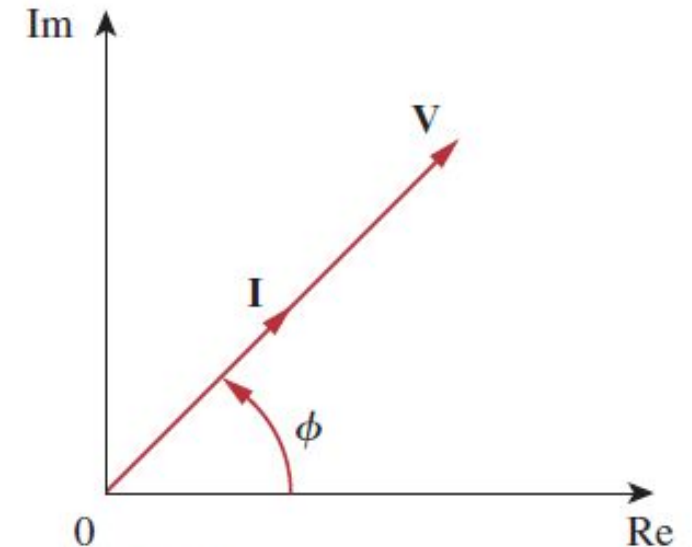


Figure 9.10
Phasor diagram for the resistor.

For the inductor L , assume the current through it is $i = I_m \cos(\omega t + \phi)$. The voltage across the inductor is

$$v = L \frac{di}{dt} = -\omega L I_m \sin(\omega t + \phi) \quad (9.32)$$

Recall from Eq. (9.10) that $-\sin A = \cos(A + 90^\circ)$. We can write the voltage as

$$v = \omega L I_m \cos(\omega t + \phi + 90^\circ) \quad (9.33)$$

which transforms to the phasor

$$\mathbf{V} = \omega L I_m e^{j(\phi + 90^\circ)} = \omega L I_m e^{j\phi} e^{j90^\circ} = \omega L I_m \angle \phi + 90^\circ \quad (9.34)$$

But $I_m \angle \phi = \mathbf{I}$, and from Eq. (9.19), $e^{j90^\circ} = j$. Thus,

$$\mathbf{V} = j\omega L \mathbf{I} \quad (9.35)$$

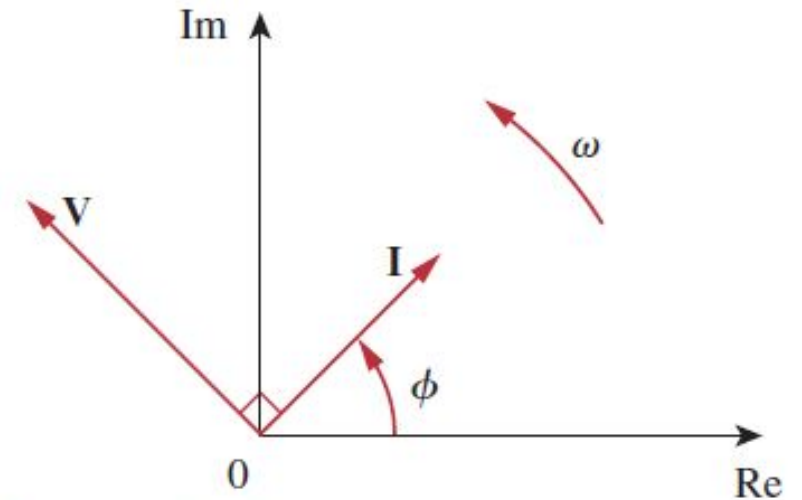


Figure 9.12

Phasor diagram for the inductor;
 $\mathbf{I}$ lags $\mathbf{V}$.

For the capacitor C , assume the voltage across it is $v = V_m \cos(\omega t + \phi)$. The current through the capacitor is

$$i = C \frac{dv}{dt} \quad (9.36)$$

By following the same steps as we took for the inductor or by applying Eq. (9.27) on Eq. (9.36), we obtain

$$\mathbf{I} = j\omega C \mathbf{V} \quad \Rightarrow \quad \mathbf{V} = \frac{\mathbf{I}}{j\omega C} \quad (9.37)$$

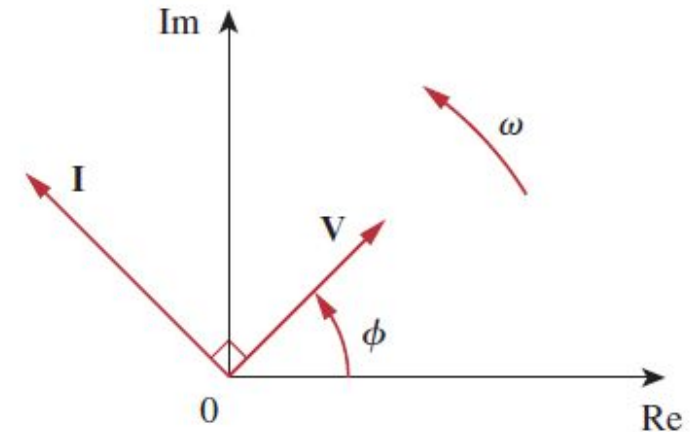


Figure 9.14
Phasor diagram for the capacitor; $\mathbf{I}$ leads $\mathbf{V}$.

TABLE 9.2

Summary of voltage-current relationships.

Element	Time domain	Frequency domain
R	$v = Ri$	$\mathbf{V} = R\mathbf{I}$
L	$v = L\frac{di}{dt}$	$\mathbf{V} = j\omega L\mathbf{I}$
C	$i = C\frac{dv}{dt}$	$\mathbf{V} = \frac{\mathbf{I}}{j\omega C}$

Example 9.8

The voltage $v = 12 \cos(60t + 45^\circ)$ is applied to a 0.1-H inductor. Find the steady-state current through the inductor.

Solution:

For the inductor, $\mathbf{V} = j\omega L\mathbf{I}$, where $\omega = 60$ rad/s and $\mathbf{V} = 12\angle 45^\circ$ V. Hence,

$$\mathbf{I} = \frac{\mathbf{V}}{j\omega L} = \frac{12\angle 45^\circ}{j60 \times 0.1} = \frac{12\angle 45^\circ}{6\angle 90^\circ} = 2\angle -45^\circ \text{ A}$$

Converting this to the time domain,

$$i(t) = 2 \cos(60t - 45^\circ) \text{ A}$$

Practice Problem 9.8

If voltage $v = 10 \cos(100t + 30^\circ)$ is applied to a $50 \mu\text{F}$ capacitor, calculate the current through the capacitor.

Answer: $50 \cos(100t + 120^\circ) \text{ mA}$.

Impedance and Admittance

In the preceding section, we obtained the voltage-current relations for the three passive elements as

$$\mathbf{V} = R\mathbf{I}, \quad \mathbf{V} = j\omega L\mathbf{I}, \quad \mathbf{V} = \frac{\mathbf{I}}{j\omega C} \quad (9.38)$$

These equations may be written in terms of the ratio of the phasor voltage to the phasor current as

$$\frac{\mathbf{V}}{\mathbf{I}} = R, \quad \frac{\mathbf{V}}{\mathbf{I}} = j\omega L, \quad \frac{\mathbf{V}}{\mathbf{I}} = \frac{1}{j\omega C} \quad (9.39)$$

From these three expressions, we obtain Ohm's law in phasor form for any type of element as

$$\mathbf{Z} = \frac{\mathbf{V}}{\mathbf{I}} \quad \text{or} \quad \mathbf{V} = \mathbf{Z}\mathbf{I} \quad (9.40)$$

where $\mathbf{Z}$ is a frequency-dependent quantity known as *impedance*, measured in ohms.

The **impedance** $\mathbf{Z}$ of a circuit is the ratio of the phasor voltage $\mathbf{V}$ to the phasor current $\mathbf{I}$, measured in ohms (Ω).

- As a complex quantity, the impedance may be expressed in rectangular form as

$$\mathbf{Z} = R + jX \quad (9.41)$$

where $R = \operatorname{Re} \mathbf{Z}$ is the *resistance* and $X = \operatorname{Im} \mathbf{Z}$ is the *reactance*.

$$\mathbf{Z} = R + jX = |\mathbf{Z}| \angle \theta \quad (9.43)$$

where

$$|\mathbf{Z}| = \sqrt{R^2 + X^2}, \quad \theta = \tan^{-1} \frac{X}{R} \quad (9.44)$$

and

$$R = |\mathbf{Z}| \cos \theta, \quad X = |\mathbf{Z}| \sin \theta \quad (9.45)$$

It is sometimes convenient to work with the reciprocal of impedance, known as *admittance*.

The **admittance** $\mathbf{Y}$ is the reciprocal of impedance, measured in siemens (S).

$$\mathbf{Y} = \frac{1}{\mathbf{Z}} = \frac{\mathbf{I}}{\mathbf{V}}$$

Find $v(t)$ and $i(t)$ in the circuit shown in Fig. 9.16.

Example 9.9

Solution:

From the voltage source $10 \cos 4t$, $\omega = 4$,

$$\mathbf{V}_s = 10 \angle 0^\circ \text{ V}$$

The impedance is

$$\mathbf{Z} = 5 + \frac{1}{j\omega C} = 5 + \frac{1}{j4 \times 0.1} = 5 - j2.5 \, \Omega$$

Hence the current

$$\begin{aligned} \mathbf{I} &= \frac{\mathbf{V}_s}{\mathbf{Z}} = \frac{10 \angle 0^\circ}{5 - j2.5} = \frac{10(5 + j2.5)}{5^2 + 2.5^2} \\ &= 1.6 + j0.8 = 1.789 \angle 26.57^\circ \text{ A} \end{aligned} \quad (9.9.1)$$

The voltage across the capacitor is

$$\begin{aligned} \mathbf{V} &= \mathbf{I} \mathbf{Z}_C = \frac{\mathbf{I}}{j\omega C} = \frac{1.789 \angle 26.57^\circ}{j4 \times 0.1} \\ &= \frac{1.789 \angle 26.57^\circ}{0.4 \angle 90^\circ} = 4.47 \angle -63.43^\circ \text{ V} \end{aligned} \quad (9.9.2)$$

Converting $\mathbf{I}$ and $\mathbf{V}$ in Eqs. (9.9.1) and (9.9.2) to the time domain, we get

$$i(t) = 1.789 \cos(4t + 26.57^\circ) \text{ A}$$

$$v(t) = 4.47 \cos(4t - 63.43^\circ) \text{ V}$$

Notice that $i(t)$ leads $v(t)$ by 90° as expected.

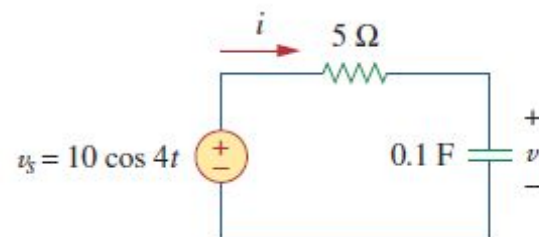


Figure 9.16

For Example 9.9.